SR-TE Yang modules

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Overview for IETF-127

Flow specification drafts

- IDR
  - Adopted – none
  - I-D: draft-wu-idr-flowspec-yang-cfg-04.txt
  - I-D: draft-lin-idr-bgp-flowspec-yang

SR structures - config

[draft-wu-idr-flowspec-yang-cfg]

Module: ietf-flowspec

(augment – missing

```
  +--rw flowspec
     +--rw flowspec-cfg
        +--rw flowspec-policy* [policy-name]
           +--rw policy-name string
           +--rw vrf-name?
           +--rw address-family
           +--rw flowspec-rule* [rule-name]
           | …
           +--rw flowspec-action
```

[draft-lin-idr-flowspec-yang]

Module: ietf-bgp-flowspec

```
 augment /rt:routing/rt:control-plane-protocols
          /rt:control-plane-protocol/bgp:bgp/bgp:global
          /bgp:afi-safis/bgp:afi-safi:
 +--rw flowspec
       +--rw prefix-limit
       |  +--rw max-prefixes?          uint32
       |  +--rw warning-threshold-pct?
                rt-types: percentage
       |  +--rw teardown?              boolean
       |  +--rw idle-time?             union
       |  +--ro prefix-limit-exceeded?  boolean
       +--rw validate-route-origin?   boolean
```

SR structures – operational state

[draft-wu-idr-flowspec-yang-cfg]
Module: ietf-flowspec
(augment – missing

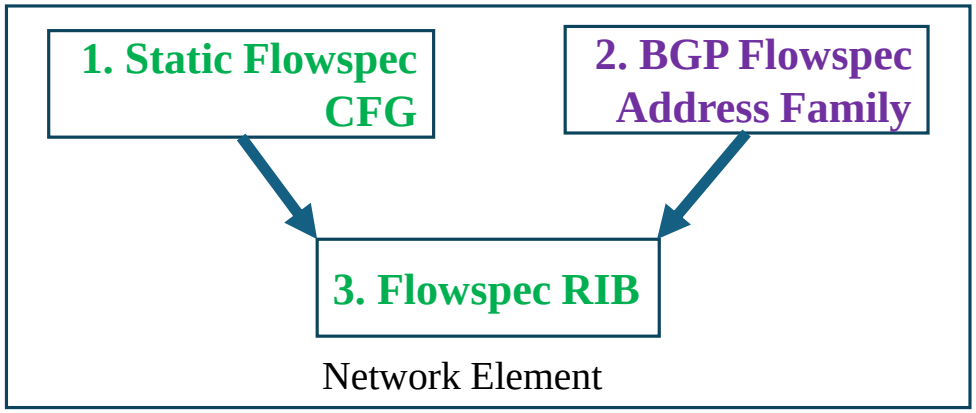
```
  +--ro flowspec-state
     +--ro flowspec-rib
     |    +--ro-flowspec-entry *
     |    | ….
     |     +--ro flowspec-protocol-specific
     |         +--ro (protocol)?
     |              +-- ro (protocol)
     |                   +--(bgp)
     |                    …. (components, actions)
     +--ro flowspec-statistics
```

[draft-lin-idr-flowspec-yang]
Module: ietf-bgp-flowspec

```
 augment /rt:routing/rt:control-plane-protocols
          /rt:control-plane-protocol/bgp:bgp/bgp:global
          /bgp:afi-safis/bgp:afi-safi:
+--rw flowspec
      +--rw prefix-limit
      |  +--rw max-prefixes?          uint32
      |  +--rw warning-threshold-pct?
               rt-types: percentage
      |  +--rw teardown?              boolean
      |  +--rw idle-time?             union
      |  +--ro prefix-limit-exceeded?  boolean
      +--rw validate-route-origin?   boolean
```

Changwang's diagram

Flowspec Structure：



1. **Static Flowspec CFG:** is used to configure static flow specification routes which contains filtering rules and corresponding actions, and send static flow specification routes to **Flowspec RIB**.
2. **BGP Flowspec Address Family:** is used to receive flow specification routes via BGP, and send flow specification routes to **Flowspec RIB**.
3. **Flowspec RIB:** is used to store flow specification routes from different protocols, such as Static protocol and BGP.

- **draft-wu-idr-flowspec-yang-cfg**

✓ Defines a YANG data model for Flow Specification implementations.

✓ This YANG model focuses on:

1. **Static Flowspec CFG**
2. **Flowspec RIB**
3. **Shared operational state for BGP - based on definitions in FS rIB**

- **draft-lin-idr-bgp-flowspec-yang**

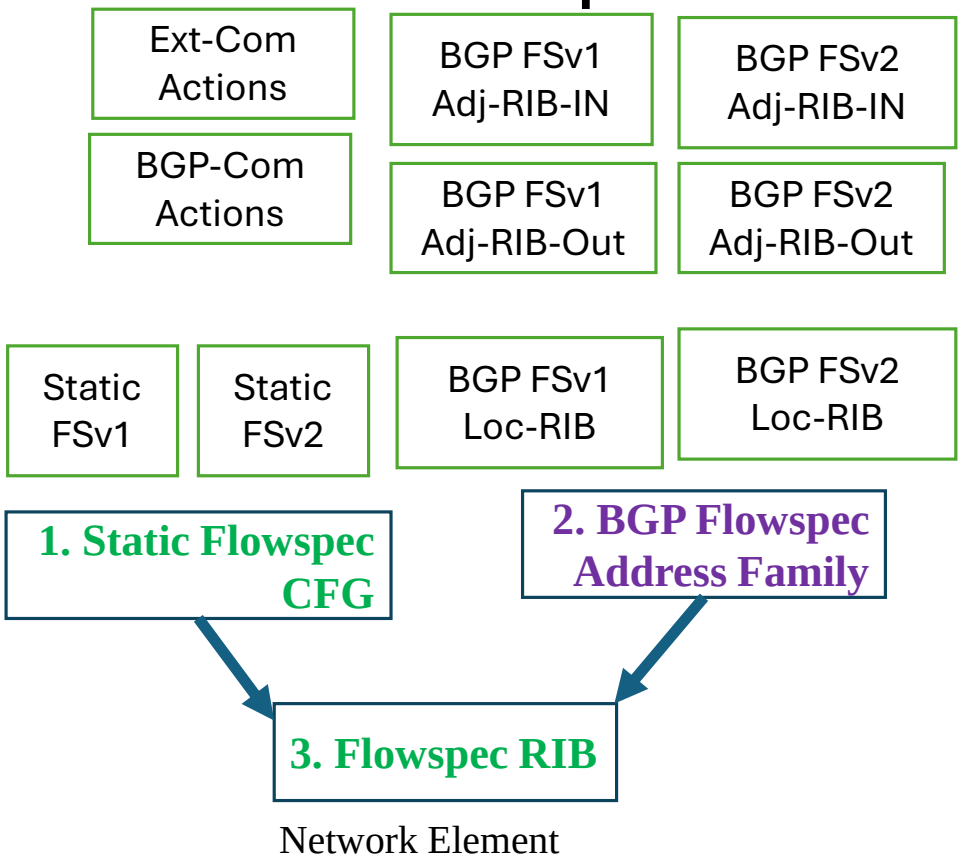
✓ Defines a YANG data model for the BGP Flowspec Address Families (IPv4 and IPv6).

✓ This YANG model focuses on:

✓ **1. BGP Flowspec Address Family**

- protocol limits and operational state

Model overlap and Issues



Issues

- Two Flow specification families
- FSv1 and FSv2
  - Each with 2 SAFIs (unicast, VPN)
  - Each Configuration, Op-State, Notifications

Flowspec needs

1. Static RIBs (Unicast, VPN) for FSv1, FSv2 and static
2. Local RIB – active FS rules
3. BGP-LOC RIB - FSv1/FSv2
4. AFI/SAFI augmentations for AdjRib
5. Links to Actions in FS format
   1) Extended Communities
   2) Attributes

Resolution

- Each model has a part of the BGP configuration
- Each model has differentiation
- Gaps exist in the models

- Two choices
  - Work to edit the modules to harden
  - Merge common configuration and operational state
- Will hold an editing session at IETF-127 – at Side-room
- Will hold an interim focused on Yang modules.